Gravitational Microlensing

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Abstract

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Introduction

The search for planets beyond our solar system is one of the most active areas of modern astrophysics. To date, over 6,000 exoplanets have been confirmed [6] using a variety of observational techniques, each sensitive to different planetary and stellar configurations. The discovery of these worlds is not only scientifically significant — it reshapes our understanding of how planetary systems form and how common they are across the galaxy — but also carries a broader human relevance, as the question of whether other planets capable of supporting life exist remains one of the most compelling open questions in science.

My interest in this topic comes from a long-standing fascination with the universe and with the idea of finding planets beyond our own solar system. Among the available detection methods, gravitational microlensing was the one that caught my attention the most. Its underlying physics — the bending of light by gravity and the way an entire planetary system can be characterized from a single brightness variation — seemed more fascinating than any other approach, and that was reason enough to choose it. The decision to combine microlensing with machine learning was driven by two reasons: a genuine desire to understand how machine learning works and what it can achieve, and a practical one — the volume of microlensing data produced by current and future surveys is too large to be analyzed manually, and automating this process is not only useful but necessary.

According to Einstein's general theory of relativity, gravity curves spacetime and therefore deflects the path of light. When a massive stellar object passes between a distant source and an observer, it acts as a gravitational lens, temporarily amplifying the observed brightness of the background source. Microlensing is a subcategory of this phenomenon where the source and lens are separated by an angular distance on the order of milli-arcseconds [5], and all physical parameters of the event are derived exclusively from the photon intensity variation of the source over time.

Gravitational lensing was first suggested under Newton's corpuscular theory of light, and later its predicted deflection angle was doubled by Einstein's general theory of relativity. Arthur Eddington confirmed this experimentally in 1919 during his solar eclipse expedition. Although Einstein himself stated in 1939 that "there is no great chance of observing this phenomenon" [2], Bohdan Paczyński revived gravitational microlensing in 1986 as an observable phenomenon in the Milky Way [1]. Initially proposed as a tool to detect dark matter concentrations, this objective was later discarded as dark matter is now known to consist of non-baryonic particles which need not be compact [3]. Today, microlensing is primarily used to detect exoplanets, though its observational difficulty is reflected in the numbers: of the 6,000+ confirmed exoplanets, only approximately 60 have been discovered via microlensing [5].

The objective of this work is to develop a complete machine learning pipeline for the automated analysis of microlensing light curves, capable of classifying events and estimating the physical parameters of the lensing system. The specific objectives are the following. The first is to develop a classifier that determines, from a light curve alone, whether a microlensing event involves a single lens or a binary lens system. The second is to develop two parameter estimation models: one for single-lens events, estimating lens mass, velocity, and distance, and one for binary-lens events, estimating all system parameters including planetary mass, orbital parameters, and distances. The third is to build an iterative optimization program that refines the parameter estimates produced by the third model with higher precision.

A central methodological challenge in this work is the scarcity of real labeled microlensing data. Microlensing events are non-repeatable — once the lens moves out of alignment, the event is gone forever — and confirmed detections of planetary systems via microlensing remain rare. This makes it impossible to train machine learning models directly on real observations. For this reason, all datasets used in this work are synthetically generated using parameter distributions derived from real observed events and published literature. Before being used for data generation, each parameter distribution is validated against real observational data: stellar mass and orbital parameter distributions are cross-checked against the NASA Exoplanet Science Institute archive [11][12], and impact parameter distributions are validated against the public OGLE-IV catalogue [9]. This ensures that the synthetic light curves are physically realistic even if not directly observed.

The methodology begins with the mathematical derivation of the microlensing equations for both single-lens and binary-lens systems. From these, the parameter distributions governing real microlensing events are characterized and validated against the observational data described above, ensuring that the synthetic datasets used to train the models are physically realistic. All code used for dataset generation and model training was written in Python, AI-assisted and subsequently revised and validated by the author.

Exoplanet detection methods

Exoplanets cannot in most cases be observed directly, as they are far too faint and too close to their host stars to be resolved from Earth. Instead, each detection method exploits a different indirect physical effect, making the various techniques complementary: each is sensitive to a different region of the planet mass versus orbital separation parameter space [5].

Radial velocity detects planets by measuring the periodic Doppler shift in the host star's spectrum induced by the gravitational pull of an orbiting planet. It is particularly effective at finding massive planets in short-period orbits, but detecting planets at orbital distances greater than 1 AU requires observing campaigns spanning years to decades, making it poorly suited to the discovery of cold, distant planets [5].

Transit photometry identifies exoplanets by detecting the periodic dimming of a star as a planet passes across its disk. It requires a precise geometric alignment between the planet's orbital plane and the line of sight, so it is inherently biased towards planets orbiting close to their host stars. It is the most prolific detection method to date, with the Kepler space mission alone having found hundreds of confirmed planets [5].

Astrometry measures extremely small periodic shifts in a star's position on the sky caused by the gravitational influence of an orbiting planet. The technique is in principle sensitive to a wide range of orbital separations, but the required positional precision is extremely demanding, and the method has only recently become viable at scale with the Gaia space mission [5].

Pulsar timing exploits the extraordinary regularity of millisecond pulsars — rapidly rotating neutron stars that emit highly stable electromagnetic pulses — to detect planets through the irregularities they introduce in pulse arrival times. It provided the first confirmed exoplanet detections in 1992, but is restricted to an exotic and rare class of stellar remnants [5].

Direct imaging is the only method that attempts to capture light emitted or reflected by the planet itself. Using adaptive optics on large telescopes, it has successfully found massive planets at very large separations from their host stars. However, because a star outshines any nearby planet by many orders of magnitude in the optical, only planets lying far from their hosts in young systems are detectable with this method [5].

Gravitational microlensing is unique among detection methods in that it requires no light to be detected from either the planet or its host star. It exploits the temporary brightening of a background source star as a foreground lens system passes in front of it, with the planet revealing itself through a brief anomaly superimposed on the main lensing event. This makes it sensitive to planets at orbital separations of roughly 1–10 AU around stars at kiloparsec distances — a regime largely inaccessible to all other methods — and it does not favor any particular type of host star [5]. Its main limitation is that events are non-repeatable and require dedicated wide-field survey programs to detect at scale.

Geometry of a microlensing system

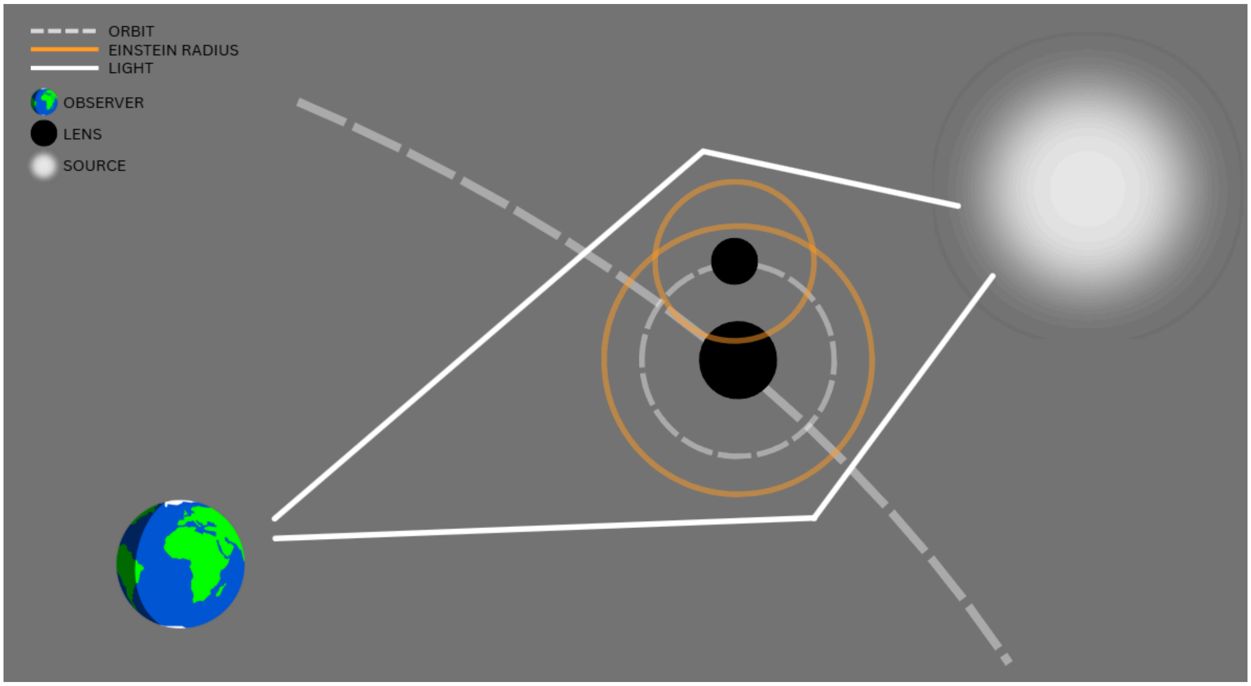

Image 1: Understanding a microlensing system's geometry. Source: Personal elaboration (with canva)

**The proportions and distances shown in the image may not be probable in reality*

A microlensing system consists of three elements: an observer (typically a telescope), a distant light-emitting source, and at least one intervening mass known as the lens, which curves spacetime and deflects the incoming light [3][7].

In practice, although all components are in motion, microlensing analyses typically assume a fixed alignment between the source and the observer while the lens transits the line of sight between them.

The observed intensity increases because the gravitational curvature deflects not only the light ray that would ordinarily reach the telescope, but also neighbouring rays, so the observer receives a higher photon flux than in the absence of the lens.

Single lens equations

The single lens equations yield a plot of photon intensity as a function of time known as the light curve, which quantifies the amplification of the source flux at each moment. The final result is Paczyński's amplification, a compact expression that nonetheless requires several intermediate quantities to be derived beforehand.

Einstein's radius

Einstein's radius is the first equation needed and it will be the most used also in the binary and N lenses equations as it represents the influence of the lens on the light deflection.

$$r_E = \sqrt{\frac{4GM}{c^2}\ \frac{D_l D_{ls}}{D_s}}$$

r_E	Radius where gravity bends light	m
G	Universal Gravitational Constant	$m^3\ kg^{-1}\ s^{-2}$
M	Lens' mass	kg
c	Light's velocity	m/s
D_{ls}	Lens - source distance	m
D_l	Lens - observer distance	m
D_s	Source - observer distance	m

Einstein's time

Einstein's time equation provides the time needed by the lens to complete 1 Einstein radius

$$t_E = \frac{r_E}{v_\perp}$$

t_E	Time for the lens to cross one Einstein radius	s
r_E	Einstein's Radius	m
$v_\perp$	Lens' velocity	m/s

Normalized lens - source separation by time

The normalized separation gives the instantaneous lens–source angular separation expressed in units of the Einstein radius.

$$u(t) = \sqrt{u_0^2 + \left(\frac{t - t_0}{t_E}\right)}$$

u(t)	Lens - source separation normalized at each moment	–
u_0	Normalized minimum separation distance	–
t	Time (in the event) where it has to be calculated	s
t_0	Second during the event with maximum alineation	s
t_E	Einstein's time	s

Paczyński's amplification

Paczyński's amplification is the final expression, giving the instantaneous magnification factor of the source flux as a function of the normalized separation

$$A(u) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}}$$

A(u)	Photon intensity increase	–
u	Normalized lens - source separation by time	–

Binary lens equations

Everything from the binary lens equations can be extended to N lenses, but as increasing the lens number by more than 2 also increases difficulty on calculus and probability of being recorded, we are going to focus just on the binary lens equations.

Binary lens equations are also derived under the thin lens approximation. They consider a primary lens, as in the single lens case, plus a secondary lens — typically a planet — orbiting the primary and producing anomalies in the single-lens light curve.

The main conceptual difference is that binary lens equations incorporate the time-varying positions of both masses, whereas the single lens case depends only on mass, distances, and relative velocity.

Masses ratio

The masses ratio, or masses fraction, is a preliminary step that, while seemingly redundant, simplifies the mass-dependent terms in all subsequent equations.

$$m_\star = \frac{M_\star}{M_\star + M_p} \qquad m_p = \frac{M_p}{M_\star + M_p} \qquad q = \frac{m_p}{m_\star}$$

$M_\star$	Main lens mass (star)	kg
M_p	Secondary lens mass (planet)	kg
$m_\star$	Star mass' percentage over the system	–
m_p	Planet mass' percentage over the system	–
q	Masses proportion	–

Einstein radius

We already know the formula for the Einstein radius from the single lens equations, but now, as we have two different lenses, we would have to calculate the radius twice. To reduce the number of calculations needed we are going to use the mass fractions (q). The planet's Einstein radius is obtained by multiplying the star's Einstein radius by √q. This follows directly from the definition, since the only difference between the two is the mass, and $r_{E(p)} = \sqrt{q} \cdot r_{E\star}$ yields the same result as computing each independently.

$$r_{E_\star} = \sqrt{\frac{4GM_\star}{c^2}\frac{D_l D_{ls}}{D_s}} \qquad r_{E_p} = r_{E_\star}\sqrt{q}$$

$r_{E(\star)}$	Radius where lens' gravity bends light	m
$r_{E(p)}$	Radius where planet's' gravity bends light	m
G	Universal Gravitational Constant	$m^3\ kg^{-1}\ s^{-2}$
M	Lens' mass	kg
c	Light's velocity	m/s
D_{ls}	Lens - source distance	m
D_l	Lens - observer distance	m
D_s	Source - observer distance	m
q	Masses fraction	–

System's Einstein angle

System's Einstein Angle represents the angle formed by the observer-source (real position) and the observer-source (observed position). As the distances involved are enormous, the Einstein angle is typically extremely small so measured in arc milliseconds (mas) instead of radians (rad). Although the Einstein angle can be expressed as $\theta_E = r_E / D_l$, the full expression is used here to make the physical dependencies explicit.

$$\theta_{E_\star} = \sqrt{\frac{4GM_{tot}}{c^2}\frac{D_{ls}}{D_l D_s}}$$

θ_E	Angle formed by observer-source (real) and observer-source (observed)	rad
G	Universal Gravitational Constant	$m^3\ kg^{-1}\ s^{-2}$
M_{tot}	System's total mass ($M_\star + M_p$)	kg
c	Light's velocity	m/s
D_{ls}	Lens - source distance	m
D_l	Lens - observer distance	m
D_s	Source - observer distance	m

Einstein's time

Einstein's time is the characteristic time for the lens to pass through one Einstein's radius.
We already know this equation from the single lens equation but now, as in the Einstein's radius, we have got to implement it also on the planet's mass.
When calculating Einstein's time for the planet the planet's orbital velocity is neglected, and the same transversal velocity as the primary lens is assumed.

$$t_{E_\star} = \frac{r_{E_\star}}{v_\perp} \qquad t_{E_p} = t_{E_\star}\sqrt{q}$$

$t_{E(\star)}$	Lens' time to complete 1 Einstein radius	s
$t_{E(p)}$	Planet's time to complete 1 Einstein radius	s
$r_{E(\star)}$	Lens' Einstein's radius	m
$v_\perp$	Lens' velocity	m/s
q	Masses fraction	–

Up to here we've defined the formulas for the main part of the binary lens understanding.

The following formulas will help define values for the after them ones which will define position to each component of the system.

Semi-major axis

The semi-major axis defines the size of the orbital ellipse described by the planet around the lens. In the special case of a circular orbit, which is rarely observed, the semi-major axis equals the orbital radius.

$$a = \frac{r_{per} + r_{aph}}{2}$$

a	Semi-major axis	m
r_{per}	Distance from the center to the perihelion of the ellipse	m
r_{aph}	Distance from the center to the aphelion of the ellipse	m

Angular velocity

Angular velocity measures how rapidly the planet orbits, in radians per second.

$$\omega = \sqrt{\frac{GM_{\star}}{a^3}}$$

ω	Planet’s angular velocity	rad/s
G	Universal Gravitational Constant	$m^3\ kg^{-1}\ s^{-2}$
$M_{\star}$	Lens’ mass	kg
a	Planet’s ellipse semi-major axis	m

Orbital angle

The orbital angle, as its name suggests, is the angle between the reference x-axis and the planet's position vector relative to the lens.

$$a(t) = a_0 + \omega\,(t - t_{ref})$$

$a(t)$	Planet's orbital angle	rad
a_0	Planet's initial angle (often defined to 0 or by the lens)	rad
ω	Angular velocity	rad/s
t	Time when the angle is calculated	s
t_{ref}	Time when $a_0 = a(t)$, (often set to 0 also)	s

Orbital time

The orbital time is a quantity that, when directly observable, allows the angular velocity to be derived straightforwardly. In practice, however, it is rarely measured directly from a microlensing event, so the angular velocity is instead computed from the physical parameters and the orbital time is derived from it.

$$T_{orb} = 2\pi/\omega$$

T_{orb}	Time for the planet to complete one lap around the star	s
ω	Planet's angular velocity	rad/s

Planet-lens distance (normalized)

The planet–lens distance varies over time, as the orbit is generally elliptical. After computing the planet–lens distance, it is normalized by the lens' Einstein radius and used in this dimensionless form in subsequent equations.

$$r(t) = \frac{a\,(1 - e^{2})}{1 + e\,cos[a(t)]} \qquad d(t) = \frac{r(t)}{r_{E_\star}}$$

d(t)	Normalized planet-lens distance	–
r(t)	Planet-lens distance	m
a	semi-major axis	m
e	Orbital eccentricity (r_{aph} - r_{per} / r_{aph} + r_{per})	(--)
a(t)	Orbital angle	rad/s
$r_{E(\star)}$	Lens' Einstein angle	m

Planet's maximum impact time

This parameter is the analogue of t0t_0 t0 for the planetary component: it gives the time at which the source passes closest to the planet's Einstein ring, producing the maximum planetary amplification.

$$t_p = t_0 + t_{E_\star} * d\,cos[a]$$

t_p	Planet's maximum impact time	s
t_0	Maximum amplification time	s
$t_{E(\star)}$	Lens' Einstein time	s
d	Planet-lens normalized distance (d(t))	–
a	Orbital Angle (a(t))	rad

Trajectory angle by time

The trajectory angle is the main determinant of how the planet's gravitational influence manifests in the observed light curve. It evolves slowly for planets with long orbital periods relative to the event duration, and remains approximately constant when the orbital time greatly exceeds the Einstein time.

$$\alpha(t) = a_{ref} - a(t)$$

α(t)	Trajectory angle by time	rad
α_{ref}	Trajectory angle at the start of the event	rad
a(t)	Planet's orbital angle	rad

Now we have finished understanding the main equations, and we are going to proceed to the positions which are going to be a fundamental part of the final amplification equation.
First of all, in microlensing, we fix the lens position to (0,0) and therefore, the source and planet are moving.

Lens position

As stated above, the lens is fixed at the origin of the reference frame, (0,0).

$$\vec{\xi}_{\star} = (0, 0)$$

Source position

The source position by the lens will be changing by time.

$$\vec{u}(t) = \left(\left(\frac{t-t_0}{t_{E_\star}}\right), u_0\right)$$

$\vec{u}$	Source position	–
t	The time when the position is calculated	s

t_0	Maximum amplification time	s
$t_{E(\star)}$	Lens Einstein time	s
u_0	Normalized minimum separation distance	–

Planet position

$$\vec{\xi}_p(t) = d(t) * (cos[a(t)],\ sin[a(t)])$$

$\vec{\xi}_p(t)$	Planet's position by time	–
d(t)	Normalized planet-lens distance	–
a(t)	Orbital angle	rad

Now we have understood all the equations needed to calculate the final amplification which is calculated by a jacobian.

Jacobian

$$J = I - \sum_{i=1}^{2} m_i \frac{\partial}{\partial \vec{\xi}} \left[\frac{\vec{u} - \vec{\xi}_i}{\left|\vec{u} - \vec{\xi}_i\right|^2} \right]$$

The full derivation of this Jacobian exceeds the scope of this work. The expression is expanded into its component form, although it is not the version implemented computationally.

In this jacobian, *i* represents the mass being currently calculated, so it has to be calculated for the main lens and also the planet.

$$J = I - \sum_{i=1}^{2} m_i \frac{1}{\left|\vec{r}_i\right|^4} \left(\frac{y_i^2 - x_i^2 \quad -2x_i y_i}{-2x_i y_i \quad x_i^2 - y_i^2} \right)$$

**Note that it is not possible to display matrix in google docs, though the terms in brackets are supposed to be a matrix*

J	The jacobian matrix	–
I	The identity matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	–
m_i	I component mass fraction	–
$\vec{r}_i$	$\vec{u} - \vec{\xi}_i$ Source-mass distance	–
x_i	X components from the r vector	–
y_i	Y components from the r vector	–

Parameters distributions

With the microlensing equations established, the next step is to generate the synthetic datasets on which the machine learning models will be trained. For the datasets to produce generalizable models, its parameters must follow distributions consistent with those observed in real events. Even though the number of confirmed microlensing detections is limited, the distributions are derived from real data available in published catalogues and from values reported in the literature, which are described individually in the subsections below.

Lens' Mass (M_⋆)

While all parameters described in this section contribute to the event, the lens mass is particularly central as it directly determines the Einstein radius.
I have extracted the data from the Nasa's exoplanet science institute where I found a table containing almost all values for a microlensing event defined from real events, real binary events. You can find the link to this table at [11].
To plot the distribution of the lens' mass I have used a python code and the results show a distribution peaking at approximately 0.45 M☉ and declining steeply beyond 0.8 M☉.

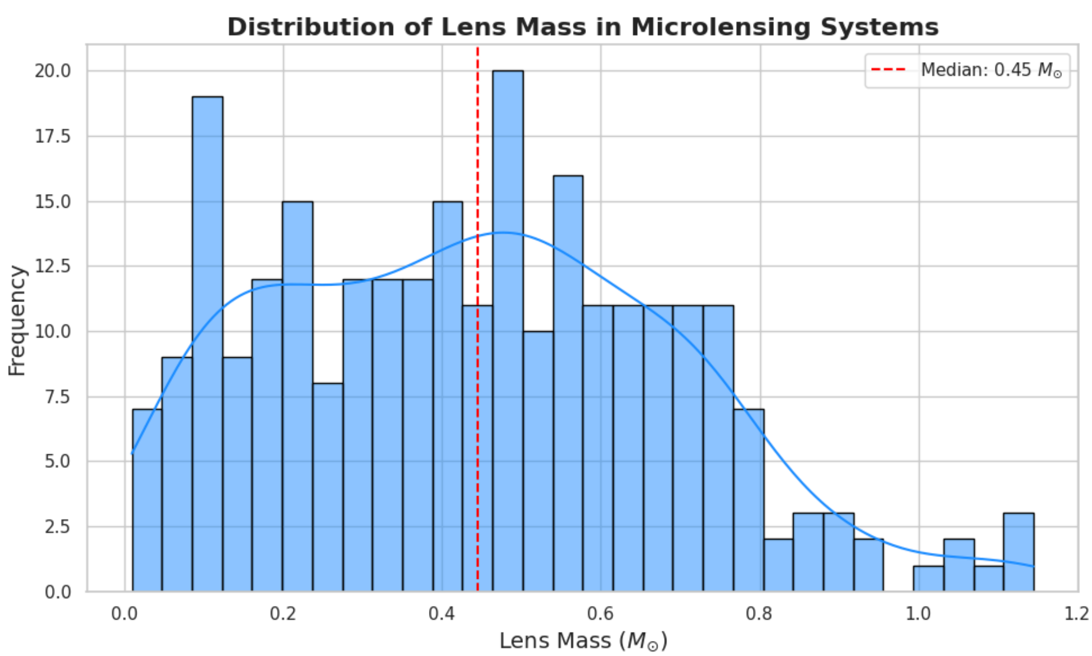

Image 2. Lens' Mass distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [11] in column Lens Mass [Solar mass]

Mass ratio (q)

The mass ratio, beyond its role in the equations, indirectly defines the bending capacity for the planet which will then alter more or less the light curve of the source.
As all the events in [11] are confirmed binary (or multiple) lens events and provide the lens mass and planet mass of the system, q is computed for each event and plotted using Python.

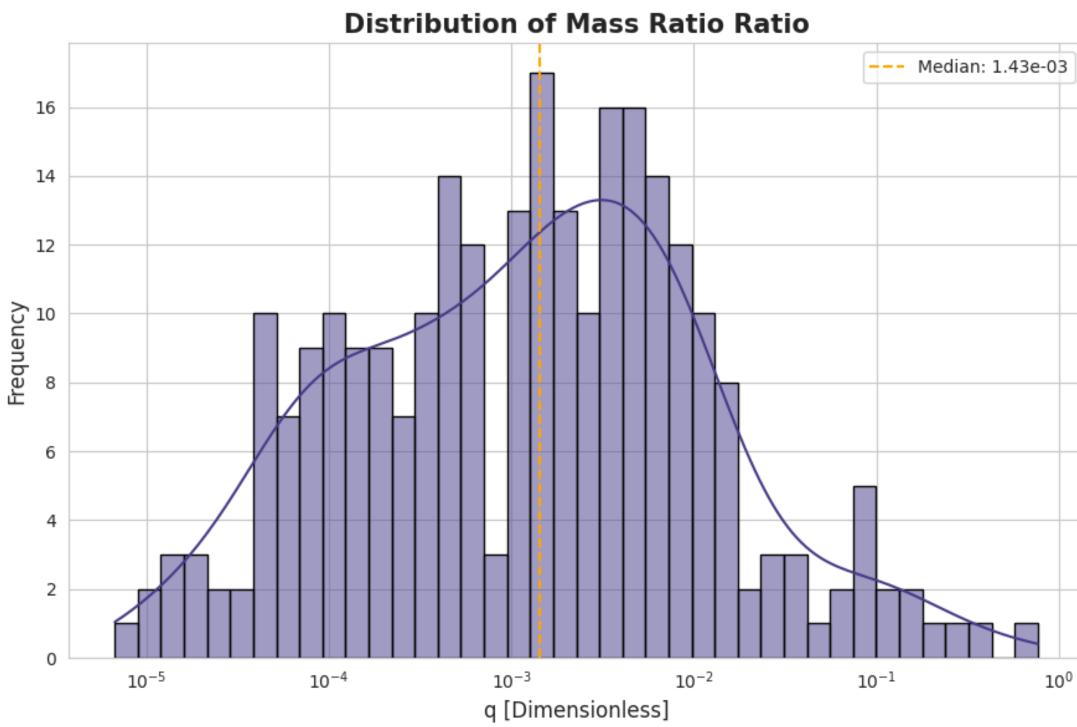

Image 3. Masses Ratio distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [11] in columns Lens Mass [Solar mass] *and* Planet Mass [Jupiter mass]

As seen on the graphic, it can be approximated as a log-normal distribution with a peak at approximately $1.43 \cdot 10^{-3}$ expressed as dimensionless mass ratio.

Distance from lens to source (s)

The distance from lens to source (named D_ls on the equations) is calculated by subtracting the distance to the lens from the distance to the source. In [11] there are only a few observations which specify the distance to source,the resulting distribution may not be fully representative;

nevertheless, it constitutes the most reliable observational reference currently available and is used accordingly. I have plotted it with python resulting in an irregular distribution which tends to 0.5 kpc but it is more or less uniform on every distance as can be seen in below:

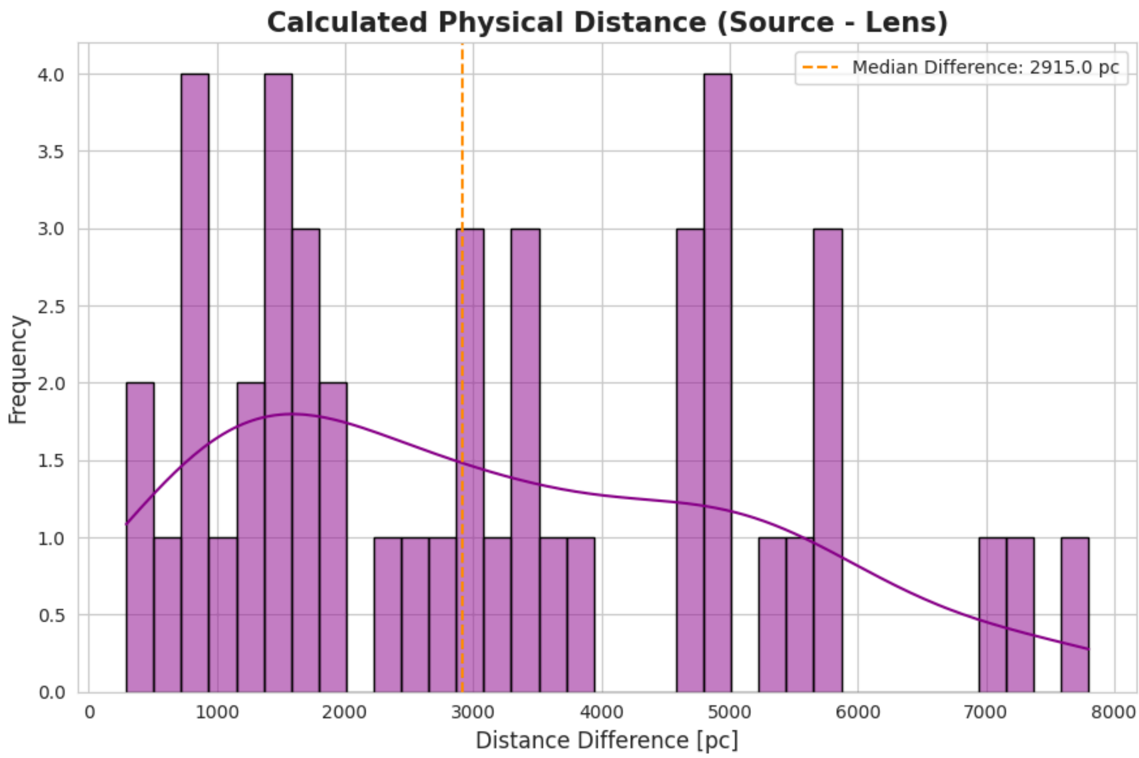

Image 4. Lens to Source Distance distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [11] in columns Lens Distance [pc] *and* Source Distance [pc]

Distance to lens (D_l)

Paczyński suggest in [10] that, as seen in Figure 4 of the paper, the distance from the observer to the lens the curve reflects a prominent peak within the bulge regime at 6 - 7 kpc, in the area where most actual telescopes look, described in the the paper as *b* which represents the angle of elevation of the observation by the center of the galactic disk, which is from 3º to 4º as there are many planets (so high microlensing event probability) but not too much galactic dust as there would be at 0º to 2º.

To ensure the validity of these distribution I checked with real data from [11] in the column *Lens Distance [pc]* by plotting it with python and the results were the following:

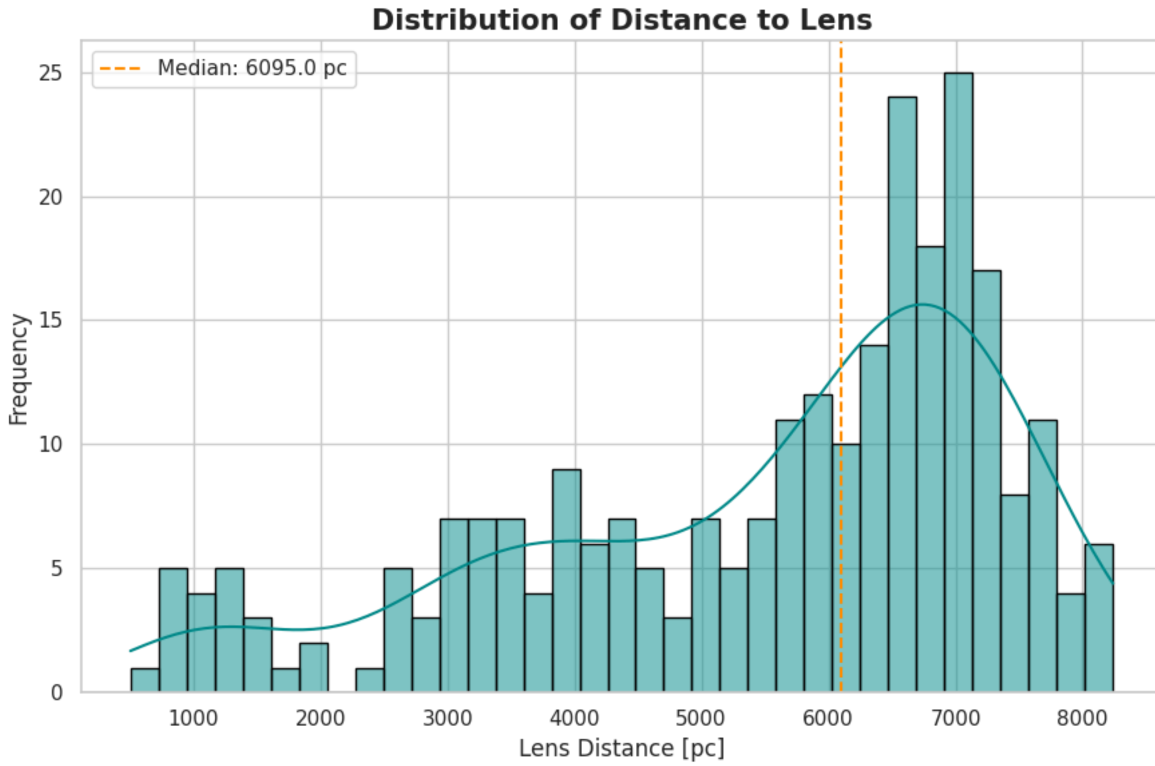

Image 5. Observer to lens distance distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [11] in column Lens Distance [pc]

In the image we can appreciate that from the data at [11] where events from OGLE, MOA, KMTNet and Kepler are published, the distribution of distance to lens is similar to what proposed by Paczyński although with the main peak at 7 pc.

Also, to see if there is any relation between the lens' mass and the distance from the observer to it I created a coverage plot for these two parameters which demonstrates that there is no apparent relation among mass and distance.

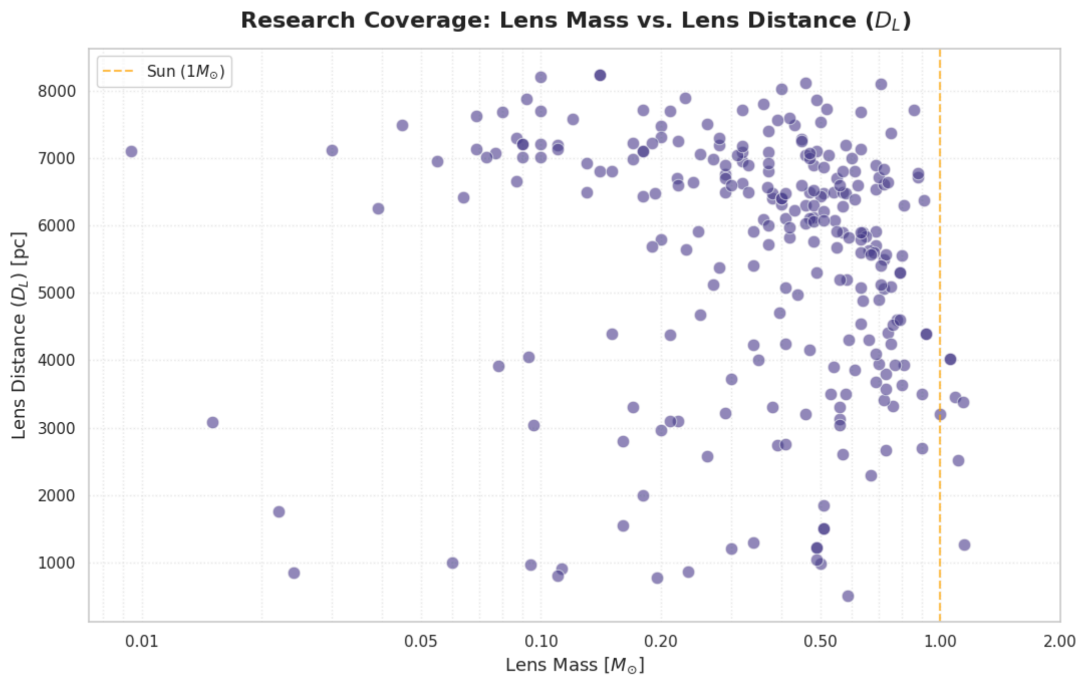

Image 6. Mass and Distance to Lens coverage.Source: Personal elaboration generated with python from data available at [11] in columns Lens Distance [pc] *and* Lens Mass [Solar mass]

Lens' velocity (v_⊥)

Lens' transversal velocity can't be measured with a simple photometry study which makes it more difficult to find real data. Therefore, a Maxwell-Boltzmann distribution is assumed, with a characteristic speed (mode) of 200 km/s, consistent with [1].

Impact parameter (u0)

Following Paczyński (1986), the impact parameter u0 is theoretically expected to follow a uniform distribution over (0,1), since the probability of a given alignment is proportional to the projected area within the Einstein ring , nevertheless, an examination of the public OGLE-IV survey data [9] resulted with the real distribution of u0 being skewed with a strong concentration near zero and a progressively decreasing tail toward larger impact parameters. This is likely an observational bias: events with u0 values close to zero produce significantly higher peak amplification and are therefore easier to detect with current telescopes.

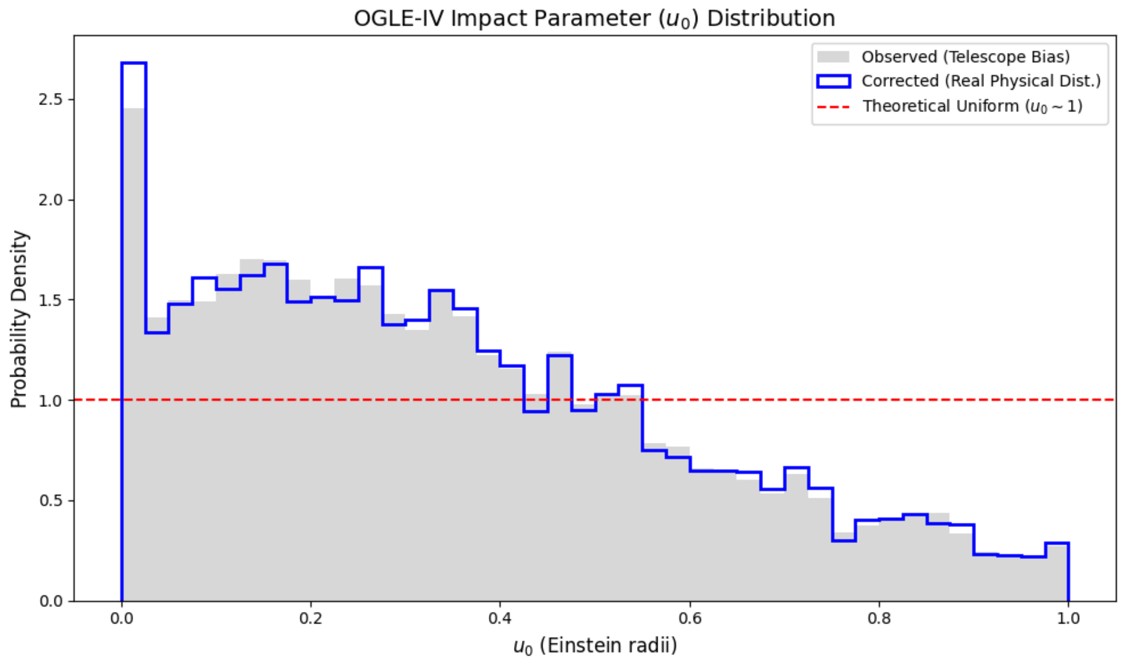

Image 7. Impact parameter distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [9]

Semi-major axis (a)

The semi-major axis is a key definer of the planet's orbit therefore important to define the planet's area of light bending.

The available data in [11] for the semi-major axis and eccentricity, which I will talk about in the next section, is limited so I looked it up on NASA's web and found [12].

In [12] you can find each parameter for every confirmed planet, not only microlensing detected ones. When opening the table you may probably wonder why there are about 40000 rows of planet data with only 6000 planets detected. This is because in that table NASA includes all publications of parameters that have been made for each planet. Filtering by the default parameter set column reduces the table to 6,128 unique planet entries (as of the time of this writing), each corresponding to the best-fitting published model for each confirmed planet.

The problem with this data is that it does not only reflect the microlensing detected parameters, which normally would be different to the ones detected by radial velocity or transit, but, for the semi-major axis, we can accept this possible error because it should theoretically be a completely random parameter which should depend, differently for the mass or distance of the lens, on the type of survey that analyzes it.

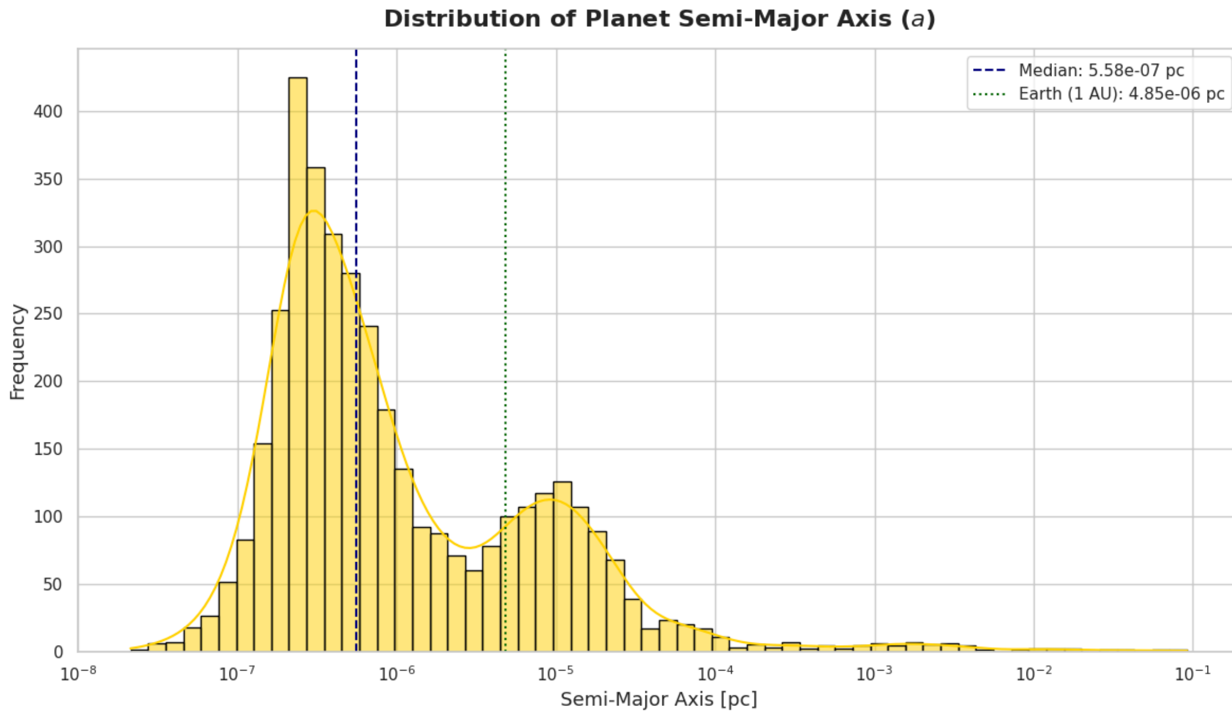

Image 8. Planet's semi-major axis distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [12] in column Orbit Semi-Major Axis [au]

In the plotted data we can see a bimodal distribution with a dominant peak between 10^{-7} and 10^{-6}, a smaller secondary bump at about 10^{-5} parsecs, and a long tail extending toward higher values.

Orbital eccentricity (e)

The orbital eccentricity, as well as the semi-major axis, describes the planet's orbit and, with these two parameters, we could know exactly the aphelion and the perihelion of each orbit.
As in with the semi-major axis data I am going to use the data found at [12] which, for the same reason, will not damage the distribution even though including events not detected via microlensing.
When analysing and plotting the data too many zeros were found, which would mean that the orbit is completely circular, and the cause was investigated. The problem is that many, usually transit, surveys assume a perfect circular orbit to make it easier to calculate any other parameter and, as, as seen in the plot below, the probably "real" eccentricity is not that far from zero, the assumed error is not that big.

In the plot, these assumed values were excluded by requiring each plotted data to also have presence in the orbital eccentricity error column, ensuring that it has been analyzed and not assumed.

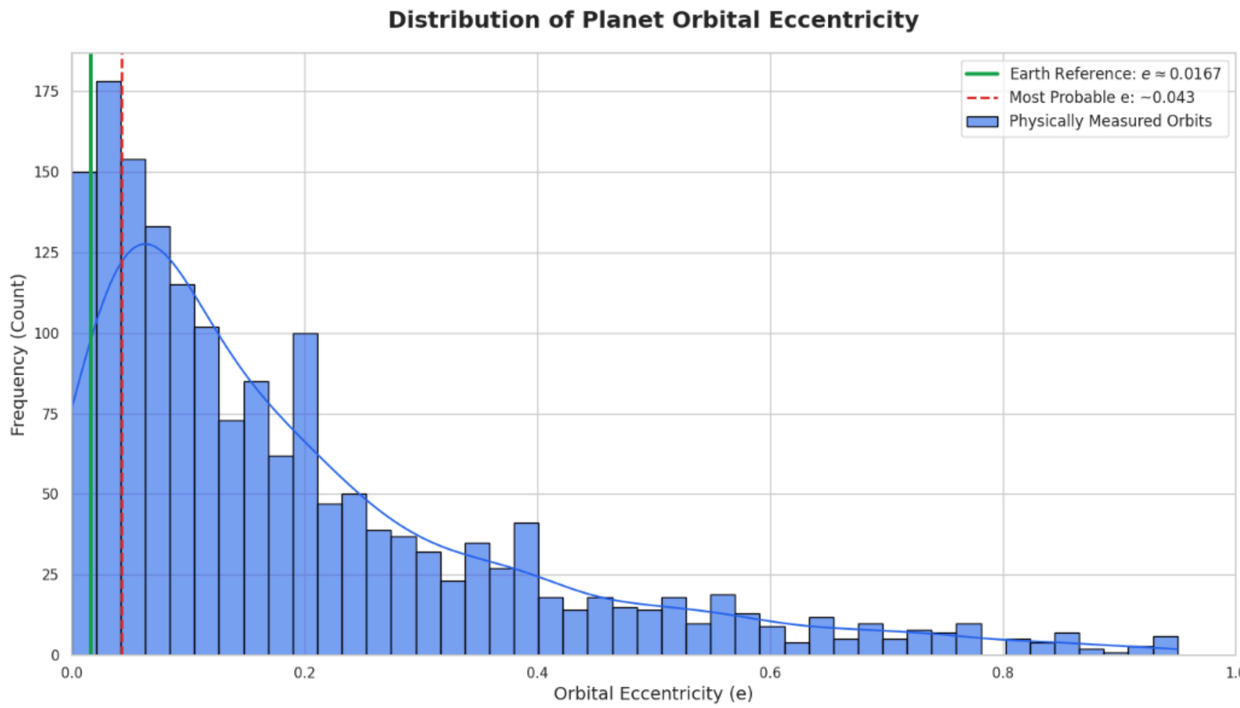

Image 9. Planet's orbital eccentricity distribution in a microlensing event. Source: Personal elaboration generated with python from data available at [12] in column Eccentricity

We could describe the plot as a long-tailed descending distribution which is probably due to the fact that older planets will always be more affected by their star's gravity and keep turning their orbit more circular and also because it is known that in systems with more than 2 or 3 planets orbits tend to be almost perfect.

Trajectory angle of reference (α_{ref})

The trajectory angle of reference is the most challenging parameter to assign when generating a binary lens dataset.

This angle defines the trajectory of the source through the lens system, which is theoretically random. However, if the source trajectory does not pass through the planet's lensing region, any of our telescopes analysing microlensing would detect the event as a single lens event (although other methods like transit or radial velocity could more easily detect the possibility of a planet in that circumstances).

The effective trajectory angle varies over time as the planet orbits the primary lens. Therefore, from a reference α (which is the one that will be randomized) its trajectory will be computed using the formula so depending mainly on the planet’s orbit.
For this reason, when generating the synthetic dataset, it is assumed that the source trajectory always passes through the planet's lensing region at some point during the event, ensuring that the planetary anomaly is detectable.

Machine Learning

To develop the final product which will predict whether a microlensing event contains a planet, more than one machine learning algorithm has to be developed. In this section, the theoretical foundations of machine learning are introduced, followed by a description of the methodology used in this work.

What machine learning is

Machine Learning is a subfield of artificial intelligence whose goal is to develop algorithms that improve their performance on a given task through experience, without being explicitly programmed for each case [14]. A standard formalization defines it as: a computer program is said to learn from experience E with respect to a task T and performance measure P, if its performance at T, as measured by P, improves with experience E [15].

Unlike traditional programming, where an expert hand-crafts a list of rules, machine learning lets large amounts of data determine the solution. This is especially valuable for tasks that are intuitive for humans but formally impossible to describe rule by rule, such as recognizing patterns in complex signals [16].

Learning tasks are generally categorized into three branches:

- Supervised learning: The model is trained on labeled examples (input-output pairs) to learn a mapping that generalizes to unseen data.
- Unsupervised learning: The model extracts hidden structure from unlabeled data, through clustering or dimensionality reduction.
- Reinforcement learning: An agent learns an optimal sequence of decisions by interacting with an environment and receiving reward or punishment signals [15].

This work exclusively uses supervised learning, as all synthetic training samples carry a known label.

The three components of machine learning

Every machine learning algorithm, regardless of its complexity, is built from three fundamental components: representation, evaluation, and optimization [15]. Understanding what each one does is key to understanding how a model actually learns.

- Representation defines the space of possible solutions the model can explore — in other words, what kind of answer it is even capable of giving. Choosing the wrong representation means the correct solution is simply unreachable, no matter how much data or computing power is available.
- Evaluation is how the model measures its own error. After making a prediction, the model needs a way to know how wrong it was and by how much. This is done through a metric.
- Optimization is the mechanism by which the model improves. Once the evaluation tells the model how wrong it is, optimization determines what to change and in which direction to reduce that error.

Together, these three components define not just how a model is built, but why it works or why it fails.

The generalization process

The ultimate goal of machine learning is generalization: the ability to perform correctly on new, unseen, data rather than only on training data [17]
Since a model cannot be evaluated on data it has seen during training, the available dataset is split into a training set and a held-out test set, typically keeping between 10% and 20% of samples for evaluation.

Successful generalization requires overcoming two main challenges: underfitting and overfitting
If the model is too simple for the data it processes the output will never be sufficiently accurate. Otherwise, if the model is too complex, it will memorize noise and spurious patterns specific to the training samples, achieving high accuracy on similar data but failing to generalize to unseen examples. [24]
Managing this balance relies on the bias-variance tradeoff, considered a universal principle in the field. Bias represents error arising from overly simplistic assumptions; high bias leads to

underfitting. Variance represents error arising from excessive sensitivity to fluctuations in the training set; high variance leads to overfitting. Optimal generalization is achieved at a moderate level of complexity where the sum of bias and variance is minimized.

Finally, the No Free Lunch Theorem establishes that no single algorithm is universally superior across all possible tasks, meaning every learner must embody specific prior knowledge or an inductive bias to generalize beyond its given data [17].

Model 1

During this section, a first machine learning model is developed with the objective of classifying gravitational microlensing events as either single-lens or binary-lens.

General

The first implementation of the model represents an idealized baseline designed for testing and validation purposes. By operating on a controlled, synthetic dataset, this section aims to verify the core methodology and understand how a classification model behaves under optimal conditions. This serves as a necessary stepping stone toward anticipating and evaluating the model's performance on real observational data.

Dataset

The dataset used to train and evaluate this model is synthetically generated, as no sufficiently large real observed dataset is publicly available with the required labelling and uniformity. All code used for dataset generation and model training was AI-assisted and subsequently revised and validated by the author.

The dataset consists of 100,000 events and 413 columns. The first column is the class label, encoding whether the event is single-lens (1) or binary-lens (2). The following 12 columns contain the physical parameters of each event, which are excluded from model input during training as they would not be available in a real observation scenario. The remaining 400 columns encode the light curve of the event.

Each light curve is centered at t_0 and spans 6 Einstein times, from $-3t_E$ to $+3t_E$. This makes it easier for the model to generalize across events with different Einstein times.

The base source intensity I_0 is fixed to 1 for all events, so that the model only receives the normalized amplification, avoiding any dependence on absolute flux that would not help the classification.

The dataset is imbalanced by design: 95,000 single-lens events and 5,000 binary-lens events, reflecting the approximate observed ratio in microlensing surveys.

The code used to generate this dataset is available at [LINK], and the dataset itself can be accessed at [LINK].

Model

Two models were trained and compared: a Random Forest and a LightGBM classifier. Both take as input the 400 light curve amplification values of each event, with the physical parameter columns excluded.

The dataset was split into 80,000 training samples and 20,000 test samples (80/20). This ratio is a standard choice in machine learning that balances having enough data to train the model and enough to evaluate it reliably. Stratified sampling was used, which ensures that the class ratio (95%/5%) is preserved in both sets, avoiding a situation where one of them ends up with a different distribution by chance.

Both models are based on decision trees and perform well on tabular data without requiring prior normalization. The Random Forest trains many independent trees and averages their results, making it robust against overfitting. LightGBM is a more efficient variant that builds trees sequentially, each one correcting the errors of the previous one. Both were trained and compared, and the one with the highest test accuracy was selected.

With 95,000 single-lens and 5,000 binary-lens events, the dataset is heavily imbalanced. A model that always predicted "single" without learning anything would already achieve 95% accuracy, making accuracy alone a misleading metric. For this reason, precision, recall, and F1-score are reported per class, as they reveal whether the model is actually learning to identify binary events or simply ignoring them. To address the imbalance during training, both models were trained with balanced class weights, which multiplies the error of each sample by a weight inversely proportional to its class frequency. This makes each misclassified binary event penalize the model 19 times more than a misclassified single-lens event, forcing the model to take the minority class seriously.

Both models achieved a test accuracy of 1.0000, with precision, recall, and F1-score of 1.0000 for both classes. The Random Forest was selected as the final model as it was the first to reach this result.

Metric	Class 1 (single)	Class 2 (binary)

Precision	1.0000	1.0000
Recall	1.0000	1.0000
F1-score	1.0000	1.0000
Accuracy	**1.0000**	

These results should be interpreted with caution. The dataset is entirely synthetic and noise-free, meaning the light curves of each class are mathematically distinct by construction, which likely explains the perfect scores. Performance on real, noisy observational data would be expected to be lower.

The code used to train the model and the model itself are available at [LINK].

OGLE-IV

This section presents the realization of the model using real observational data tailored specifically for the OGLE-IV survey. While the previous iteration demonstrated the theoretical viability of the classification pipeline under perfect conditions, this model is developed to confront the realities of actual astronomical observations. The primary objective shifts from an idealized proof of concept to a practical, robust deployment capable of handling the inherent complexities of real-world stellar monitoring—such as observational noise, irregular sampling, and blending. By training and optimizing the architecture on actual OGLE-IV light curves, this model is designed to deliver realistic performance metrics, understand how the pipeline adapts to genuine survey conditions, and serve as an active tool for automated event classification in modern microlensing surveys.

Dataset

The dataset used to train the OGLE-IV prepared model will be similar to the general one but real OGLE-IV noise and cadence will be added so that the model learns to process

Conclusions

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